

Solution to Problem: Voltage Instability in Power Systems

Based on ELEC0447 - TP6

November 27, 2025

1 Derivation of the Relationship between Load Active Power P and Bus 2 Voltage magnitude V

The complex power S consumed by the load at Bus 2 is given by:

$$S = P + jQ = \overline{V}I^*$$

The current flowing from Bus 1 to Bus 2 is $I = \frac{\overline{E} - \overline{V}}{jX}$. Thus, the complex power S at Bus 2 is:

$$S = \overline{V} \left(\frac{\overline{E} - \overline{V}}{jX} \right)^*$$

Substituting the phasor expressions $\overline{E} = E$ and $\overline{V} = V \angle \delta$:

$$S = (V \angle \delta) \left(\frac{E - V \angle \delta}{jX} \right)^*$$

Using the property $\left(\frac{1}{j}\right)^* = (-j)^* = j$:

$$S = V \angle \delta \left(-j \frac{E - V \angle \delta}{X} \right)^* = V \angle \delta \left(j \frac{E - V \angle (-\delta)}{X} \right)$$

$$S = \frac{V}{X} \angle \delta (jE - jV(\cos \delta - j \sin \delta))$$

$$S = \frac{V}{X} \angle \delta (jE - jV \cos \delta - V \sin \delta)$$

$$S = \frac{V}{X} (\cos \delta + j \sin \delta) (-V \sin \delta + j(E - V \cos \delta))$$

By expanding $S = P + jQ$ and separating the real and imaginary components:

Active Power P

$$P = \text{Re}(S) = \frac{V}{X} [\cos \delta (-V \sin \delta) + \sin \delta (E - V \cos \delta)]$$

$$P = \frac{V}{X} [-V \cos \delta \sin \delta + E \sin \delta - V \sin \delta \cos \delta]$$

$$P = \frac{EV}{X} \sin \delta \tag{1}$$

Reactive Power Q

$$Q = \text{Im}(S) = \frac{V}{X} [\cos \delta (E - V \cos \delta) - \sin \delta (-V \sin \delta)]$$

$$Q = \frac{V}{X} [E \cos \delta - V \cos^2 \delta + V \sin^2 \delta]$$

$$Q = \frac{V}{X} (E \cos \delta - V) \tag{2}$$

Eliminating δ

We solve equations 1 and 2 for $\sin \delta$ and $\cos \delta$:

$$\sin \delta = \frac{PX}{EV}$$
$$\cos \delta = \frac{QX + V^2}{EV}$$

Using the identity $\sin^2 \delta + \cos^2 \delta = 1$:

$$\left(\frac{PX}{EV}\right)^2 + \left(\frac{QX + V^2}{EV}\right)^2 = 1$$

Multiplying by $(EV)^2$:

$$P^2 X^2 + (QX + V^2)^2 = E^2 V^2$$

Expanding the second term:

$$P^2 X^2 + Q^2 X^2 + 2QXV^2 + V^4 = E^2 V^2$$

Rearranging into a quadratic equation in V^2 :

$$V^4 + (2QX - E^2)V^2 + (P^2 X^2 + Q^2 X^2) = 0$$

Solving for V^2 using the quadratic formula:

$$V^2 = \frac{-(2QX - E^2) \pm \sqrt{(2QX - E^2)^2 - 4(P^2 X^2 + Q^2 X^2)}}{2}$$
$$V^2 = \frac{E^2 - 2QX \pm \sqrt{4Q^2 X^2 - 4E^2 QX + E^4 - 4P^2 X^2 - 4Q^2 X^2}}{2}$$
$$V^2 = \frac{E^2}{2} - QX \pm \sqrt{\frac{E^4}{4} - E^2 QX - P^2 X^2}$$

Finally, taking the square root to find V :

$$V = \sqrt{\frac{E^2}{2} - QX \pm \sqrt{\frac{E^4}{4} - E^2 QX - P^2 X^2}}$$

If the load power factor angle is ϕ , where $Q = P \tan \phi$, the relationship is:

$$V = \sqrt{\frac{E^2}{2} - XQ \pm \sqrt{\frac{E^4}{4} - X^2 P^2 - E^2 XQ}}$$

2 Maximum Transmissible Power $P_{\max}$

The maximum power transmissible occurs at the point of voltage instability, where the two solutions for V^2 merge, meaning the discriminant (Δ) of the quadratic equation for V^2 is zero:

$$\Delta = \frac{E^4}{4} - E^2 QX - P^2 X^2 = 0$$

Substitute $Q = P \tan \phi$:

$$\frac{E^4}{4} - E^2 X(P_{\max} \tan \phi) - P_{\max}^2 X^2 = 0$$

Multiplying by -4 :

$$4P_{\max}^2 X^2 + 4E^2 X(\tan \phi)P_{\max} - E^4 = 0$$

Dividing by $4X^2$:

$$P_{\max}^2 + \left(\frac{E^2 \tan \phi}{X}\right)P_{\max} - \left(\frac{E^4}{4X^2}\right) = 0$$

Using the quadratic formula for $P_{\max}$ and selecting the positive root:

$$P_{\max} = \frac{-\left(\frac{E^2 \tan \phi}{X}\right) + \sqrt{\left(\frac{E^2 \tan \phi}{X}\right)^2 - 4(1)\left(-\frac{E^4}{4X^2}\right)}}{2}$$

$$P_{\max} = \frac{1}{2} \left[-\frac{E^2 \tan \phi}{X} + \sqrt{\frac{E^4 \tan^2 \phi}{X^2} + \frac{E^4}{X^2}} \right]$$

$$P_{\max} = \frac{E^2}{2X} \left[-\tan \phi + \sqrt{\tan^2 \phi + 1} \right]$$

Using $\tan^2 \phi + 1 = \sec^2 \phi = 1/\cos^2 \phi$:

$$P_{\max} = \frac{E^2}{2X} \left[-\tan \phi + \frac{1}{\cos \phi} \right]$$

Substituting $\tan \phi = \sin \phi / \cos \phi$:

$$P_{\max} = \frac{E^2}{2X} \left[\frac{1}{\cos \phi} - \frac{\sin \phi}{\cos \phi} \right]$$

$$P_{\max} = \frac{E^2}{2X} \left(\frac{1 - \sin \phi}{\cos \phi} \right) \quad (3)$$

Important Parameters

The important parameters for increasing the transmissible power $P_{\max}$ are:

1. **Bus 1 Voltage (E):** $P_{\max}$ is proportional to the square of the sending-end voltage (E^2).
2. **Line Reactance (X):** $P_{\max}$ is inversely proportional to the line reactance ($1/X$).
3. **Load Power Factor ($\cos \phi$):** The maximum power is dependent on the load's power factor angle ϕ . A more capacitive load (negative ϕ) allows for a higher $P_{\max}$ compared to an inductive load (positive ϕ).